Analysing The Customer Sentiments Towards Hybrid Cars

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ABSTRACT

The prices of fossil fuels used in vehicles are increasing day by day along with that there is a growing concern towards environment safety. People have started looking for more sustainable options. This research was intended to find out the view of customers of hybrid cars in India. Techniques like reviewing previous literature and using machine learning algorithms to do the analysis of online reviews of hybrid vehicles were done. Environmental safety, cost effectiveness, and societal views were the influencing factors for them to purchase hybrid cars. In spite of all these, the consumers of hybrid vehicles in India care mostly about the amount they spend, the quality, and the performance of these vehicles. The availability of charging places was also a concern. Two thousand online reviews Indian hybrid car user reviews were taken and most of them were positive suggesting a positive sentiment towards hybrid cars. With the sentimental analysis and word cloud analysis, consumer perception could be found out and they include quality, efficiency, safety, and comfort. This information can be useful for companies as well as other stakeholders in making consumers purchase hybrid cars in India and adopt them better.

INTRODUCTION

The Indian transport industry faces a critical turning point and faces several barriers in terms of growing environmental concerns. These problems also include an increase in the cost of gallons of petrol and it highlights how important it is to switch to sustainable mobility solutions. Hybrid electric vehicles (HEVs) with their unique feature of the combination of electric motors and internal combustion engines (ICE), present a stimulating blend between fully electric vehicles and conventional petrol or diesel engine cars. Before HEVs accomplish their maximum effectiveness in the Indian market, it is fundamental to figure out and debunk consumer prejudices about these luxurious vehicles. Multiple factors affect consumer's assessments of hybrid electric cars.

Further, Indian customers become more conscious about the environment as a result of air pollution and climate change. Electric and hybrid vehicles are considered to be more environmentally friendly because of their lower carbon emissions and they cause less environmental effects. Boosting the use of HEVs may promote the adoption of HEVs. The rate of usage of HEVs is still low when compared to petrol or diesel-engine cars, even though programs like the Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME) plan (Ministry of Heavy Industries and Public Enterprises, 2023), developed to promote the usage of HEVs. It would be essential to have a thorough insight into customer views to fill the adoption barrier.

Several approaches would be needed to conduct a thorough investigation into Indian consumers' opinions regarding hybrid electric cars. The utilization of a research approach that integrates quantitative surveys, qualitative focus groups, and sentiment analysis of Internet data is required, as per the findings of Srinivasan et al. (2021) and Raghuram (2019). The Indian automotive industry's stakeholders can create focused campaigns to encourage the use of hybrid electric vehicles by utilizing the information obtained from extensive customer research. Though quantitative surveys can provide researchers with systematic data on consumer preferences, qualitative focus groups provide a deeper understanding of the underlying concerns and variables impacting consumer behavior. Trends and consumer moods can be obtained through the examination of internet data, such as social media exchanges and customer reviews, mood.

In a nutshell, comprehending and regulating consumer attitudes towards HEVs is essential to promoting their wider adoption in the Indian market. By adopting an integrated study strategy and tailoring tactics to client needs, stakeholders may fully realize the promise of HEVs as a sustainable mobility option in India's evolving transportation sector. This study will look at the following research issues to find out how customers feel about HEVs:

What opinions do consumers have regarding hybrid cars based on what they've read online?

· Which specific features make hybrid automobiles the most desirable and important to consumers?

Literature Review

There is a growing need for protecting the environment and practicing sustainability. This trend has led to people looking for more eco-friendly options while selecting cars. Fossil fuel-powered cars need an alternative and one such alternative is hybrid cars. Unlike Electric vehicles, hybrid cars require more awareness and it is crucial to understand the sentiments of existing customers to fill the gap. Also, this will help manufacturers, policymakers, and other stakeholders to make better decisions regarding the vehicle. This literature review helps us to understand the past research work done and sentimental analysis techniques helping future research.

Consumer Purchase Motivations and Decision-Making

The concern for the environment, eco-friendliness, and societal views are all the main factors in selecting hybrid cars (Lee, 2011). Consumers looking for more eco-friendly options better than fossil fuel cars are attracted to hybrid cars because they contribute less towards environmental pollution (Schuhwerk et al., 2013). Fuel efficiency is a driving factor for consumers who look for ccost-effective options. Societal influence plays a vital role in influencing the buying nature of consumers (Jones et al., 2019). Many consumers have their own set of beliefs and perceptions regarding how their cars should be and that also forms as a basis for the adoption of hybrid vehicles (Brown et al., 2018).Besides these consumers feel emotionally satisfied when they give back to the environment by being more careful towards it and this adds up value to their purchase (Smith & Johnson, 2015). Societal image develops as we start using more environment friendly products. Along with this brand image and views on the utility of the product will greatly influence the buying intention of prospective consumers (Zhang et al., 2018).

Consumer Perceptions and Attitudes

Insights and conclusions regarding consumer perception and attitudes can be found out by analyzing online hybrid car reviews. Techniques such as Machine Learning and Natural Language Processing in these online reviews will help them to understand their current pain points and concerns (Pang & Lee, 2008). A deeper understanding of the consumer's preference can be easily found out as well as the future area of development and new technological advancements helping to solve the current concerns can be devised. Feedback helps to find the emotions of the users. Sentimental analyses have been done previously by conducting much research on electric vehicles and so it will be applicable in the case of hybrid cars also (Wang et al., 2020). The manufacturers and the policymakers can use these reviews to help make better decisions regarding manufacturing or any other process connected to it.

There are alsways mixed reviews when it comes to consumer perceptions and atitudes. Some may be interested in the product the others may not be (Brown et al., 2018). Some the common issues that comes to the minds of cinsumers are the initial cost of buying , the availability of charging stations, the operating difference between traditional gasoline powered cars and electric ranges. (Goodwin et al., 2019; Sierzch et al., 2019). To address these concerns manufacturers and stakeholders can device and bring out solutions and assurance to mitigate these concerns as far as possible.

Sentiment Analysis of Online Reviews: Unlocking Customer Voices

Consumer Learning and Adoption Patterns: A Historical Perspective

It is extremely important to know what is the consumer's learnings from using hybrid cars and that can be used to increase the adoption rate. Hybrid vehicles have been available for a long time but the increase in adoption has been brought out lately in the 90s (Green & White, 2017). The constant increases in fuel prices have led to customers more interest in hybrid cars and electric vehicles (EPA, 2021).

With the increasing environmental pollution and its related consequences, the government has introduced many initiatives to facilitate potential consumers to buy environmentally friendly cars.

More research in this area can help in educating future consumers of vehicles to adopt environmentally friendly alternatives to traditional fossil fuel-powered cars.

Reference	**Theme of the study**	**Method**	**Key Findings**
Lee, J. (2011)	Factors influencing the adoption of hybrid electric vehicles: A consumer behavior perspective	Consumer behavior analysis	Environmental concerns, economic benefits, and social influence are key motivators for HEV purchase intent.
Schuhwerk, M. E., Chang, S. E., & Droste, N. (2013)	Consumer perspectives on hybrid electric vehicles: A segmentation analysis	Segmentation analysis	Identified distinct consumer segments based on attitudes towards HEVs.
Jones, C. R., Bengler, K., Drechsler, K., & Ruff, J. (2019)	Influence of consumer knowledge on perceived electric vehicle characteristics	Survey and analysis	Consumer knowledge impacts perceptions of electric vehicle attributes
Brown, A., Sathaye, N., Balaji, M., & Krishna, G. (2018)	Social influence and consumer behaviour in the context of electric vehicles: Insights from qualitative studies and choice experiments	Qualitative studies and choice experiments	Social influence plays a significant role in consumer behaviour towards EVs.
Smith, R., & Johnson, D. (2015)	Consumer learning and hybrid vehicles	Research analysis	Consumer learning influences adoption patterns of hybrid vehicles.
Zhang, Y., Liu, Y., & Mao, Y. (2018)	Effect of car brand reputation on purchase intention: A moderating role of product involvement	Survey and moderation analysis	Car brand reputation moderates the relationship between product involvement and purchase intention.

Goodwin, P., Fielding, G. J., & Pearce, J. M. (2019)	Effects of latent factors and observable factors on the adoption of electric vehicles and hybrid electric vehicles in US states	Statistical analysis	Latent and observable factors impact EV and HEV adoption in US states.
Sierzch, A., Bohrer, A., & Jochem, P. (2019)	Influence of electric vehicle charging infrastructure on market diffusion: Lessons learned from the Northeastern United States	Case study analysis	Charging infrastructure significantly influences market diffusion of electric vehicles.
Pang, B., & Lee, L. (2008)	Opinion mining and sentiment analysis	Literature review and analysis	Overview of sentiment analysis techniques for mining opinions from text data.
Wang, Y., Chai, Y., Zhao, Y., Hu, J., & Cao, G. (2020)	Hybrid model for sentiment analysis of reviews for new energy vehicles	Machine learning approach	Developed a novel hybrid model for sentiment analysis of new energy vehicle reviews.
Green, E. H., & White, J. S. (2017)	Development of hybrid electric vehicles: A history perspective	Historical analysis	Explored the historical development of hybrid electric vehicles from a technological perspective.
U.S. Environmental Protection Agency (EPA). (2021)	Federal and state incentives for electric vehicles and charging infrastructure	Policy review	Overviewed federal and state incentives for EVs and charging infrastructure in the U.S.

Methodology

The data is collected from popular Indian hybrid car review websites available online. A total number of two thousand reviews were gathered from these websites for our analysis. The reviews of the customers were taken along with the ratings (5 - excellent, 4 - very good, 3 - good, 2 - poor, and 1 - very poor). This data undergoes various steps such as data pre-processing, Construction process, and Sentiment analysis, World cloud displays the most prominent words in data and classification with different ML algorithms and supervised learning algorithms for classification, such as Naive Bayes and Support Vector Machine.

DATA COLLECTION

We gathered a dataset of 2000 user reviews from a website that offers reviews on hybrid vehicles. Customers' experiences with the hybrid cars are reflected in the ratings that accompany these reviews, which range from 1 to 5. The collection includes positive and negative reviews, providing information about several facets of the car models being examined. This dataset is useful for learning about consumer attitudes and preferences in the car sector.

SENTIMENTAL ANALYSIS

Using the Python NLTK package, sentiment analysis has been done on the text reviews. A Python module called NLTK keeps up with Python's development to meet the demands of researchers and educators in computational linguistics and natural language processing.

DATASET GENERATION MODULE

PRE-PROCESSING

Cleaning and preparing the dataset for analysis was part of the data pre-processing step. This phase of sentiment analysis is crucial since it increases precision and syntactic correction. Pre-processing the data involves removing special characters, stemming words, turning uppercase to lowercase, and digitally removing words that don't communicate any feeling

TRAINING A MODEL

Scikit-learn provides easy-to-implement machine learning models with minimal code, such as Support Vector Machines. First, check if NumPy is installed correctly before using the needed model from sci-kit-learn. After training, use the same instance for testing and save the results.

Naïve Bayes and Support Vector Machine

The Bayesian network classifier, incorporating the simple Bayes, is a popular supervised classification method. It assigns words positive or negative probabilities and classifies data probabilistically using the Bayes theorem. It calculates each word's possibilities in a phrase and selects the word with the highest chance, to determine the text's mood. This method analyzes two classes—positive and negative—without directly analyzing documents and generates binary classification results.

Conversely, support vector machines (SVM) are used to determine the optimal surface for differentiating between positive and negative training data. In training and classifying text for sentiment polarity modeling, SVM goes further than traditional x/y predictions. Instead, it links text data characteristics (X) with either positive or negative tags (Y). The classifier is taught to produce a positive or negative emotion, representing a binary decision. By mapping input vectors to binary output values (0 or 1), SVM functions employ an input-output format, defining a vector space for input and a binary sentiment output.

WORD CLOUD GENERATION

A word cloud visually displays recurring and connected concepts. These clouds provide clear, concise visual data that often encourages further research. The NLTK program builds word clouds to show commonly used terms in positive and negative evaluations.

Result

Besides ratings from Indian hybrid car review websites, data was also collected from user reviews online. The study gathered 2000 reviews and ratings, with scores ranging from 1 (very poor) to 5 (excellent). Negative reviews accounted for 89.1% of the total (10%). This section presents the results of sentiment analysis using the Naïve Bayes (NB) and Support Vector Machine (SVM) classification models on reviews by hybrid car owners. The models' ability to distinguish between positive and negative user comments is thoroughly assessed using classification reports. Word cloud visualizations enhance analysis by revealing trends in machine learning algorithms' assessment of user perceptions and experiences with fast lending apps. These insights guide future research and optimize decision-making systems. The objectives of this comprehensive approach include: * Evaluating model performance: Assessing the strengths and weaknesses of Naive Bayes (NB) and Support Vector Machines (SVM) models in classifying sentiment in user reviews of hybrid cars.

Recognize user perspectives: To better understand customer experiences, find reoccurring themes and difficulties mentioned in reviews—both favorable and negative.

Examine the elements that affect decisions: Examine the fundamental causes behind people's decision to use hybrid cars.

Classification

Table 1: Classification report of naïve Bayes

	precision	recall	f1-score	support
Positive	**0.31**	**0.12**	**0.17**	**67**
Negative	**0.92**	**0.98**	**0.95**	**733**
Macro Avg	**0.62**	**0.55**	**0.56**	**800**

A substantial class imbalance is shown while interpreting the classification report for a 90% overall accuracy Naive Bayes model, which has 733 negative cases and just 67 positive ones. With an F1-score of 0.17, the positive class has low precision (31%), indicating many false positive predictions, and low recall (12%), showing the model misses a large fraction of true positives. On the other hand, the negative class exhibits good precision (92%), recall (98%), and F1-score (0.95), indicating successful negative occurrence identification. Although macro-averaged metrics offer a broad overview, the imbalance may cause them to misrepresent performance for the minority positive class. The model does a good job of identifying negative examples, but it has trouble predicting positive classes, even with its high overall accuracy. Cost-conscious education and data balancing techniques like oversampling or undersampling could address these challenges, emphasizing the importance of exploring such strategies further

Classification report of SVM.

	precision	recall	f1-score	support
Positive	**0.41**	**0.13**	**0.20**	**67**
Negative	**0.93**	**0.98**	**0.95**	**733**
Macro Avg	**0.67**	**0.56**	**0.58**	**800**

An SVM model's classification report shows a 91% overall accuracy, which is a strong result. While the model correctly identifies negative examples (high precision), it struggles to identify positive ones (low precision of 41%). This results in many false positives. Additionally, the model misses a significant proportion of true positive examples (low recall of 13%). Its F1-score (0.20) is also low, indicating weak performance in finding true positive examples. This bias towards the negative class is likely due to the imbalanced dataset with more negative examples. Even though the model is generally accurate, the uneven distribution of classes (class imbalance) can skew the results. It's crucial to consider the consequences of misidentifying positive cases. To improve performance, techniques like adjusting parameters and resampling data can be employed. Despite its overall accuracy, the model's limited ability to detect positive cases requires further analysis and improvement.

4.2 Confusion Matrix

Confusion Matrix for Naïve Bayes (NB):

	Predicted Positive	**Predicted Negative**
Actual Positive	715	18
Actual Negative	59	8

Confusion Matrix for Support Vector Machine (SVM):

	Predicted Positive	**Predicted Negative**
Actual Positive	720	13
Actual Negative	58	9

The confusion matrices of the Naive Bayes (NB) and Support Vector Machine (SVM) models compare predicted classes to actual classes. In the NB model, 715 out of 733 positive cases were correctly predicted, while in the SVM model, 720 out of 733 positive cases were correctly predicted. Both models performed better at predicting positive events. However, there is room for improvement, particularly in increasing the true positive rate

4.2.1 Sensitivity and Specificity

Sensitivity

Naïve Bayes:

Sensitivity (True Positive Rate) for Positive class:

Sensitivity = True Positives (TP)/ True Positives (TP) + False Negatives (FN)

Sensitivity = 715 / (715 + 59)

Sensitivity ≈ 92.3%

With a high sensitivity of 92.3 percent, the model accurately detects a sizable number (almost 92 percent) of real positive instances. This is consistent with the confusion matrix's large number of True Positives (715).

Support Vector Machine:

Sensitivity = True Positives (TP)/True Positives (TP) + False Negatives (FN)

Sensitivity = 715 / (715 + 59)

Sensitivity ≈ 92.4%

With a high sensitivity of 92.4 percent, the model accurately detects a substantial percentage (almost 92 percent) of real positive instances.

Specificity

Naïve Bayes:

Specificity (True Negative Rate) for Negative class:

Specificity = True Negatives (TN)/True Negatives (TN) +False Positives (FP)

Specificity = 8 / (8 + 18)

Specificity ≈ 30.7%

Nevertheless, the specificity (30.7%) is really poor. This suggests that the model has difficulty differentiating between genuine negatives and false positives. This low specificity is partly due to the large number of False Positives (18) relative to True Negatives (8).

Support Vector Machine:

Specificity for Negative class:

Specificity = True Negatives (TN)/True Negatives (TN) + False Positives (FP)

Specificity = 8 / (8 + 18)

Specificity ≈ 30.8%

But the specificity is just 30.8%, which is very poor. This suggests that the model has difficulty differentiating between genuine negatives and false positives.

4.2.2 Accuracy

Naïve Bayes (NB):

True Positives (TP) = 715

True Negatives (TN) = 8

Accuracy = (715 + 8) / 800 = 0.904 or 90.4%

Support Vector Machine (SVM):

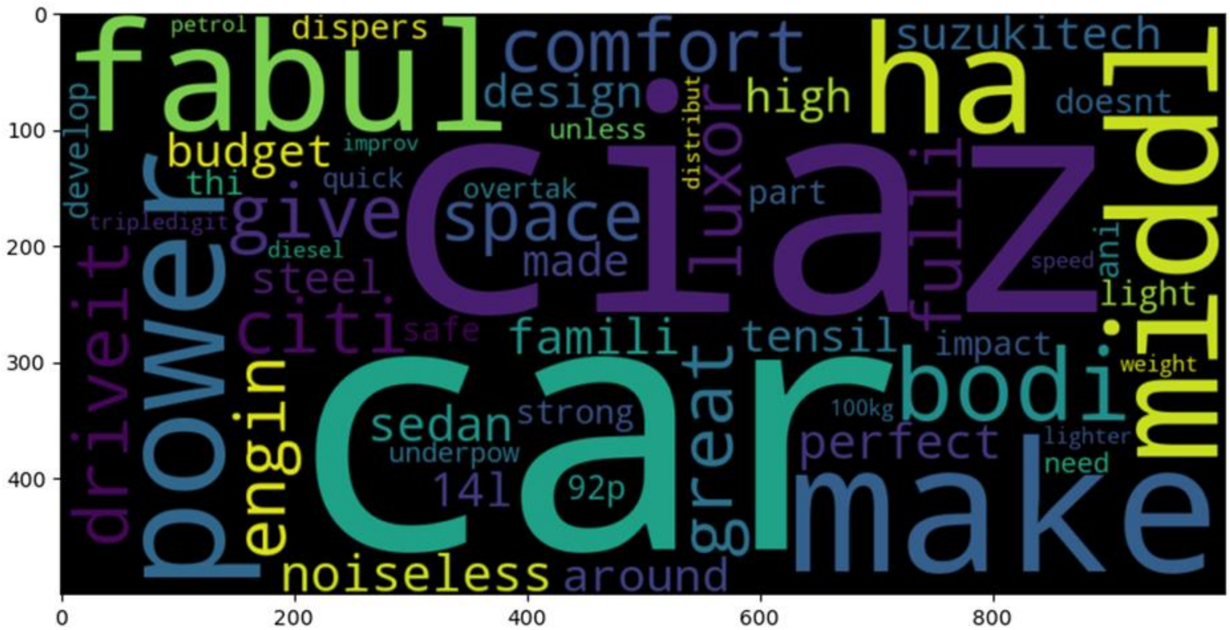
True Positives (TP) = 720

True Negatives (TN) = 9

Accuracy = (720 + 9) / 800 = 0.906 or 90.6%

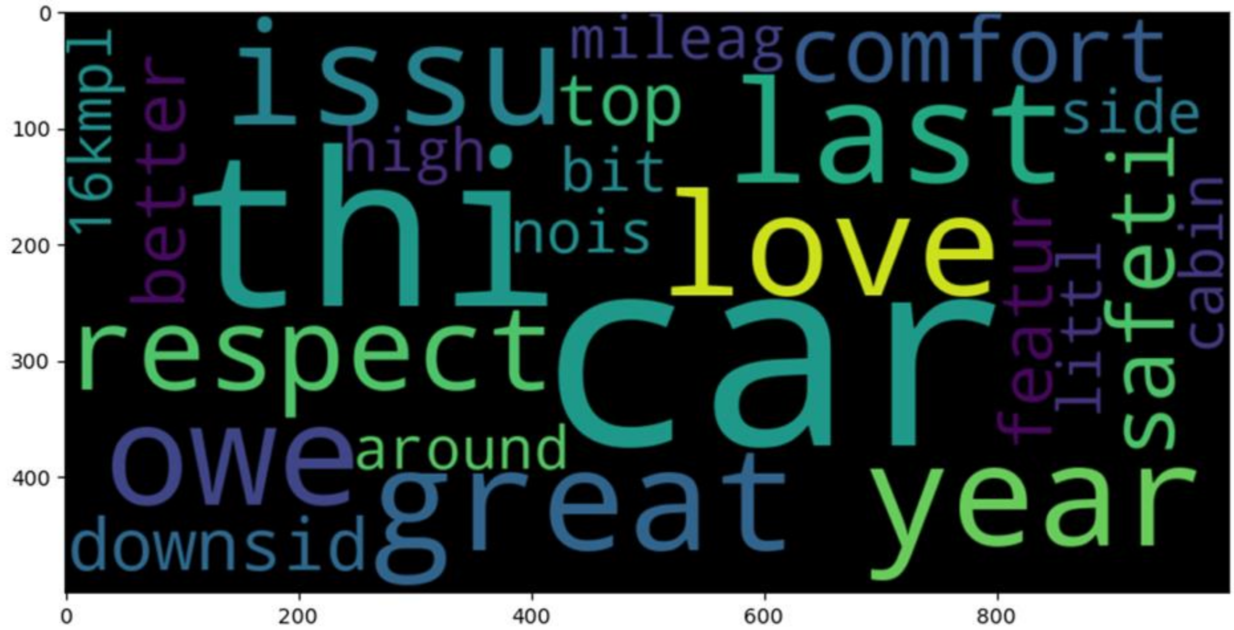
Both Naive Bayes (90.4%) and Support Vector Machine (SVM) (90.6%) show good performance in identifying positive and negative instances based on their accuracy ratings. This indicates that the great majority (more than 90%) of the data items within each category were accurately identified by them.Although the SVM model outperforms Naive Bayes in terms of accuracy (0.2%), the difference is not very great. It's crucial to keep in mind that a variation this little could not be statistically significant. Furthermore, these accuracy levels may vary depending on the particular dataset that is used.

Word Cloud from Text Analysis



Positive word cloud from text analysis

The terms that appear most frequently in customer reviews of hybrid cars are shown by the word cloud. It proposes conversations on different car makes and models, focusing on performance elements like horsepower and engine efficiency in addition to comfort and design elements like quietness and spacious interiors. Affordability and technical developments are among the economic and environmental aspects that are emphasized. Customer assessments take into account safety factors like structure and safety features. Reviews on hybrid automobiles generally indicate that people are satisfied with them, highlighting the benefits of performance, comfort, economy, and safety. This study highlights the key variables that have a substantial impact on consumers' perceptions of hybrid cars, providing producers and prospective purchasers with insightful information**.**



Negative word cloud from text analysis

The thoughts and subjects from consumer reviews of hybrid cars are probably reflected in the word cloud. Bigger terms like "love" and "great" convey good emotions and a sense of general contentment. Words like "mileage," "comfort," and "safety" probably stand for important features that buyers consider while evaluating hybrid vehicles. Furthermore, terms like "higher," "better," and "feature" could refer to customer comparisons of features and performance. But phrases like "issue," "noise," and "downside" imply that the reviews contain complaints or issues. It's important to realize that although the word cloud offers an overview of recurring topics, a detailed analysis of the actual feedback is necessary to fully comprehend client perspectives.

5. Discussion

A range of approaches were used in the study to assess consumer attitudes on hybrid cars, including word clouds and machine learning-based sentiment analysis in addition to the collecting of data from internet reviews. Its primary goal was to gather additional information about consumer attitudes and preferences about hybrid cars as well as the variables influencing these opinions.

1. Understanding Consumer Sentiments:

An examination of 2000 user reviews allowed for a more sophisticated understanding of consumer perceptions regarding hybrid cars. The dataset represented a balanced view of client attitudes by including both positive and negative sentiments. However, there was a clear bias in favor of negative reviews, indicating that consumers had significant misgivings or complaints about hybrid vehicles.

2. Key Attributes Valued by Customers:

The study included sentiment analysis and word cloud visualizations to identify the most prevalent themes and characteristics that consumers of hybrid cars valued**.**. Positive opinions were frequently focused on safety, comfort, performance, and fuel economy. Features like horsepower, fuel efficiency, roomy cabins, and general economy were important to customers. On the other hand, unfavorable opinions centered on problems like noise and the supposed drawbacks of hybrid technology.

3. Comparison with Existing Literature:

The study's findings are consistent with current literature on variables impacting customer views of hybrid electric cars. Motivators such as environmental concerns, economic gains, and social influence were consistent with previous study. However, specific issues mentioned in customer evaluations, such as performance and infrastructural restrictions, helped to deepen our understanding of consumer attitudes toward hybrid automobiles.

4. Implications for Stakeholders:

Manufacturers, legislators, and marketers all benefit from understanding customer opinion. The study findings include specific recommendations for increasing product features, resolving consumer problems, and developing focused marketing methods. Manufacturers, for example, may prioritize performance and price while investing in charging infrastructure to increase customer trust and promote wider adoption of hybrid vehicles.

5. Future Research Directions:

The report creates the foundation for upcoming studies on consumers' perceptions of hybrid cars. Future studies could examine additional factors like brand reputation and after-sales support that affect consumer impressions. Studies with a longitudinal design, which track changes in consumer views over time, may provide insightful information about how well marketing campaigns and legislative initiatives work to increase the adoption of hybrid electric vehicles.

In summary, the survey offers valuable insights into the perceptions of Indian consumers regarding hybrid automobiles. To determine the main factors influencing consumer impressions, the study analyzes internet reviews and sentiment data. The results provide useful recommendations for interested parties to encourage the broader use of hybrid vehicles in India's transportation industry.

6. Conclusion

In summary research, on how consumers view hybrid cars offers insights into the preferences of the market. By analyzing reviews and sentiment tools the study uncovers both negative aspects of customer attitudes towards hybrid vehicles.

The findings highlight the importance of addressing customer concerns while emphasizing the benefits of cars such as their performance, comfort, fuel efficiency, and safety. To promote the adoption of cars in India stakeholders, like manufacturers politicians, and marketers can tailor their strategies based on understanding these sentiments. Further research can examine additional factors that influence consumer opinions in greater detail and track changes in opinions over time. Thanks to this ongoing research, stakeholders will be able to respond to customer needs and promote the adoption of sustainable mobility solutions by the Indian market.

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